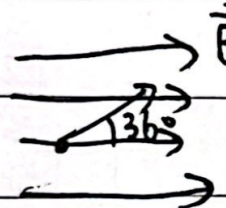


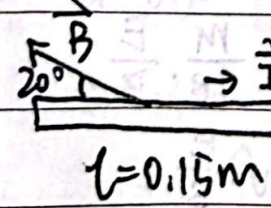
27.1.2  $\vec{F}_B = q\vec{v} \times \vec{B}$ $F_B = qvB \sin \theta$, $v = 1.95 \times 10^5 \text{ m/s}$, $B = 0.75 \text{ T}$
 $q = \frac{F_B}{vB \sin \theta} = 2.91 \times 10^{-15} \text{ C}$

28.33. $d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{l} \times \vec{r}}{r^2}$ $d\vec{l} \perp \vec{r}$
 $= \frac{\mu_0}{4\pi} \frac{Idl}{r^2}$ I direction: into the page

$B = \int dB = \frac{\mu_0}{4\pi} \frac{I(\pi R)}{R^2} = \frac{\mu_0 I}{4R}$

$B_{\text{total}} = \frac{\mu_0}{4\pi} (I_1 - I_2)$ direction: into the page

If $I_1 = I_2 \Rightarrow B = 0$

27.7.1  $I = 3.5 \text{ A}$, $B = 0.0136 \text{ T}$
 $\vec{F} = I\vec{l} \times \vec{B}$

$= IlB \sin 160^\circ = 2.44 \times 10^{-3} \text{ N } \hat{k}$

$\hookrightarrow +z$ direction



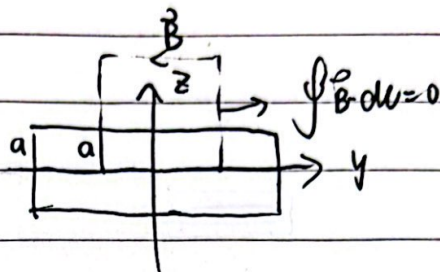
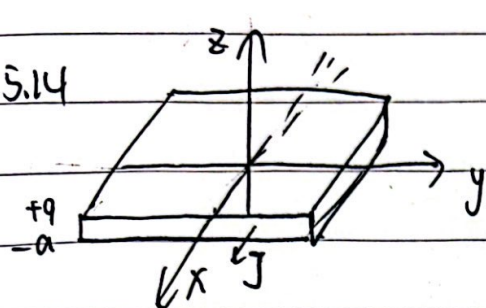
5.15



1) inside 2) between 3) outside

1) $\mu_0 |n_1 - n_2| I$ 2) $\mu_0 n_2 I$ 3) 0

5.14



infinitely large.

when $z > 0$. direction: $-\hat{y}$, when $z < 0$. direction: $+\hat{y}$ When $z = 0$. $B = 0$. (symmetry, trade off)1° $|z| < a$. $B \cdot l = |B| l = 1 |z| \cdot J \mu_0 \Rightarrow |B| = \mu_0 J |z|$, $\vec{B} = -\mu_0 J z \hat{y}$ 2° $|z| > a$. $B \cdot l = l a J \mu_0 \Rightarrow |B| = \mu_0 J a$ $\vec{B} = \begin{cases} -\mu_0 J a \hat{y}, & z > a \\ \mu_0 J a \hat{y}, & z < -a \end{cases}$

5.11

 $d\vec{l} = n \hat{x} dx$

$$B = \int \frac{\mu_0 a^2}{2(x^2 + a^2)^{3/2}} d\vec{l}$$

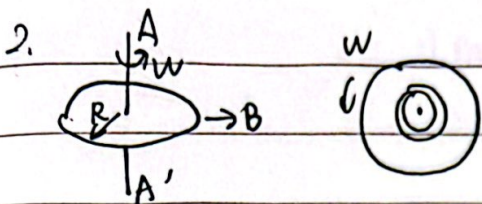
$$= \frac{\mu_0 n I}{2} \int_{x_1}^{x_2} \frac{a^2}{(a^2 + x^2)^{3/2}} dx \quad x = a \cot \theta$$

$$dx = -\frac{a}{\sin^2 \theta} d\theta, \quad \frac{1}{(a^2 + x^2)^{3/2}} = \frac{\sin^3 \theta}{a^3}$$

$$B = \frac{\mu_0 n I}{2} \int_{\theta_1}^{\theta_2} a^2 \cdot \frac{\sin^3 \theta}{a^3} \left(-\frac{a}{\sin^2 \theta}\right) d\theta$$

$$= -\frac{\mu_0 n I}{2} \int_{\theta_1}^{\theta_2} \sin \theta d\theta = \frac{\mu_0 n I}{2} (\cos \theta_2 - \cos \theta_1)$$





$$\Delta I = 2\pi r \Delta r \cdot \sigma \frac{\omega}{2\pi} = 2\pi \Delta r \cdot \sigma \cdot \frac{\omega}{2\pi} = \omega r \Delta r \cdot \sigma$$

$$\Delta m = \Delta I \cdot S = \omega r \sigma \Delta r \cdot \pi r^2 \rightarrow \text{magnetic dipole moment}$$

$$m = \int dm = \omega \sigma \pi \int_0^R r^3 dr = \frac{1}{4} \pi \omega \sigma R^4$$

1. x x x x

x x x x

a.e x x x x

x x x x

$$F = qVB = m \frac{v^2}{R} \Rightarrow R = \frac{mv}{qB}$$

$$T = \frac{2\pi R}{v} = \frac{2\pi m}{qB}$$

1° $F = mg$ $qV_0 B = mg$ $V_0 = \frac{mg}{qB}$

2° $F \neq mg$ $V_0 = V_1 + V_2$

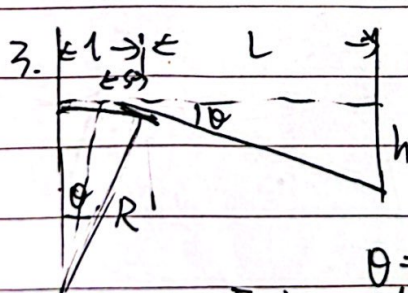
uniform motion circular motion

$$V_1 = \frac{mg}{qB}, \quad \frac{S}{V_1} = N T, \quad T = \frac{2\pi m}{qB}$$

$$S = V_1 \cdot N T = \frac{mg}{qB} \cdot N \cdot \frac{2\pi m}{qB} = N \cdot \frac{2\pi m^2 g}{q^2 B^2}, \quad N = 1, 2, 3, \dots$$

if $S \neq N \cdot \frac{2\pi m^2 g}{q^2 B^2}$ ($N = 1, 2, 3$), $V_0 = \frac{mg}{qB}$

if $S = N \cdot \frac{2\pi m^2 g}{q^2 B^2}$, no matter how large is V_0 , it pass b.



$$\tan \theta = \frac{h}{L+S}$$

$$1-S = R \tan \frac{\theta}{2}, \quad 1 = R \sin \theta$$

$$\theta \rightarrow \tan \theta \approx \theta, \quad \sin \theta \approx \theta$$

$$\Rightarrow \theta = \frac{h}{L+S}, \quad 1-S = \frac{R}{2} \theta, \quad 1 = R \theta \rightarrow \theta = \frac{1}{R}, \quad S = \frac{1}{2} L$$

$$\theta = \frac{h}{L+\frac{1}{2}L}$$

$$R = \frac{1}{\theta} = \frac{1}{h} (L + \frac{1}{2}L) = \frac{mv}{qB} = \frac{m}{qB} \cdot \frac{E}{B}$$

$$\frac{q}{m} = \frac{E}{B^2} \cdot \frac{1}{R} = \frac{E}{B^2} \cdot \frac{1}{L + \frac{1}{2}L}$$

(equilibrium: $qVB = qE \Rightarrow v = \frac{E}{B}$)

